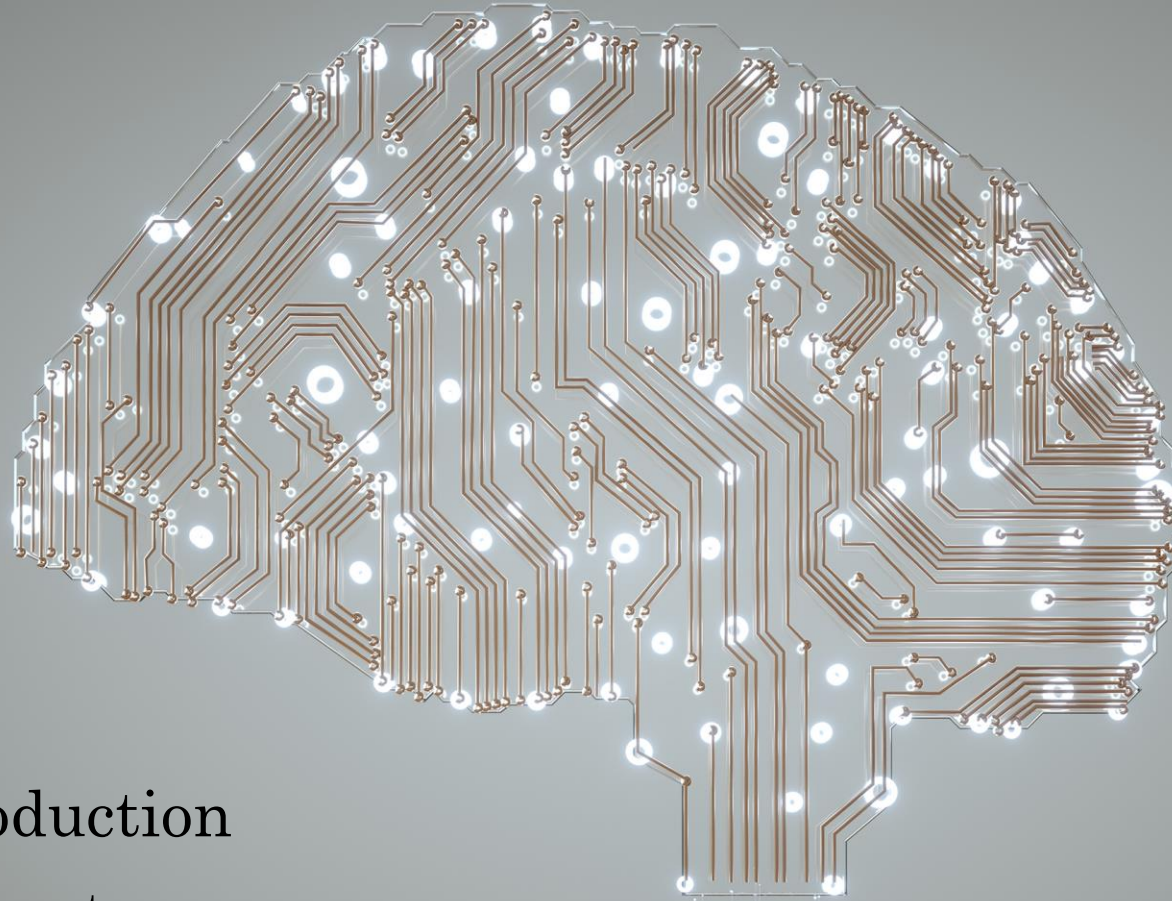


ML & NLP: PGR210



Course Introduction

Dr. Arvind Keprate

Learning Outcomes

- No Mandatory Assignment.
- 4-week individual home exam.
- Grad Scale: A to E.

Læringsutbytte

Knowledge

The student...

- can understand the basic data structures and algorithms for machine learning
- knows the mathematical concepts underlying the design and analysis of machine learning techniques appropriate for a given data science problem
- has insights into the strengths and weaknesses of Dimensionality Reduction Algorithms: variance thresholds, correlation thresholds, principal component analysis (PCA), linear discriminant analysis (LDA)
- can explain the basic natural language processing concepts and techniques for text analytics
- knows the text analytics methods and tools (such as text classification, semantic textual similarity, neural language models etc) for various data science domains

Skills

The student...

- can select appropriate machine learning methods (such as linear models, classification models, text classification, semantic textual similarity, and neural language models) and tools for a given data science problem
- can analyze mathematically the performance of machine learning methods and techniques
- can apply techniques from the course to new data science problems in terms of selection of appropriate machine learning methods, techniques and tools
- can use python or similar to implement machine learning methods and techniques

General competence

The student...

- can differentiate the suitability and efficiency of programs in terms of the machine learning methods and techniques (incl. text analytics) employed
- can apply the knowledge of and skills in machine learning in various data science domains
- can critically reflect on the tradeoffs in the design and implementation of machine learning methods and techniques

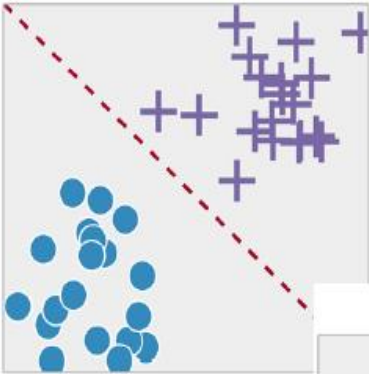
Main Topic



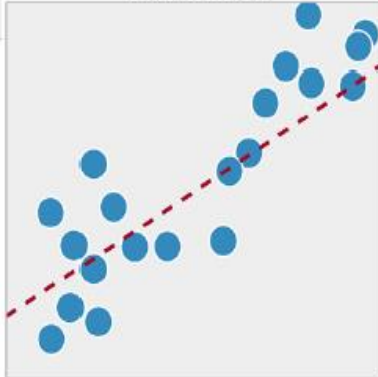
Machine Learning

1. Supervised Learning

Classification

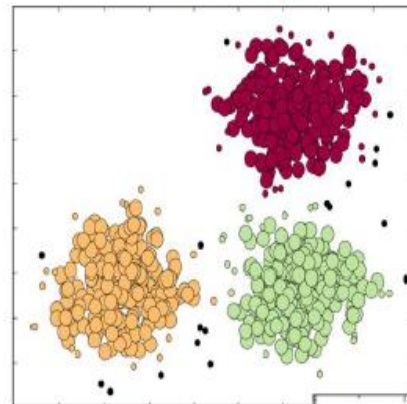


Regression

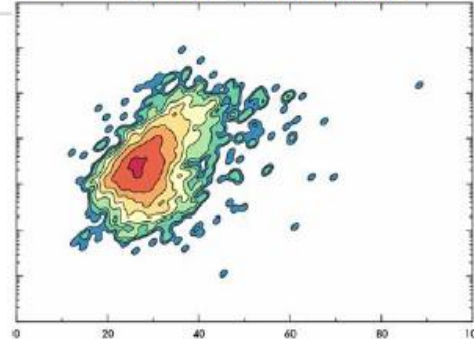


2. Unsupervised Learning

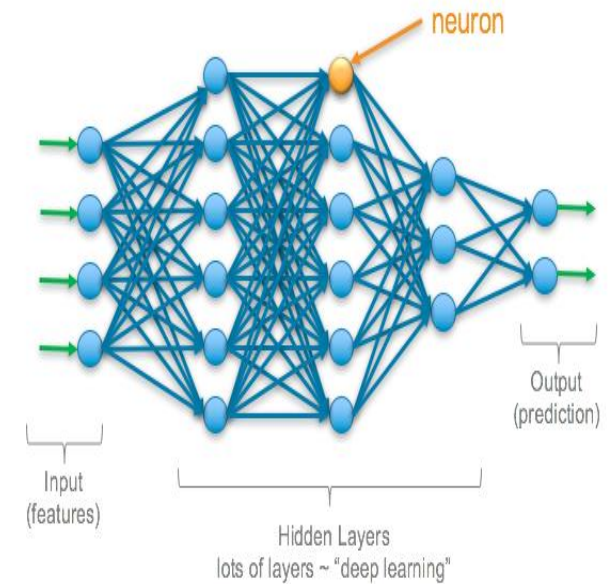
Clustering



Density Estimation

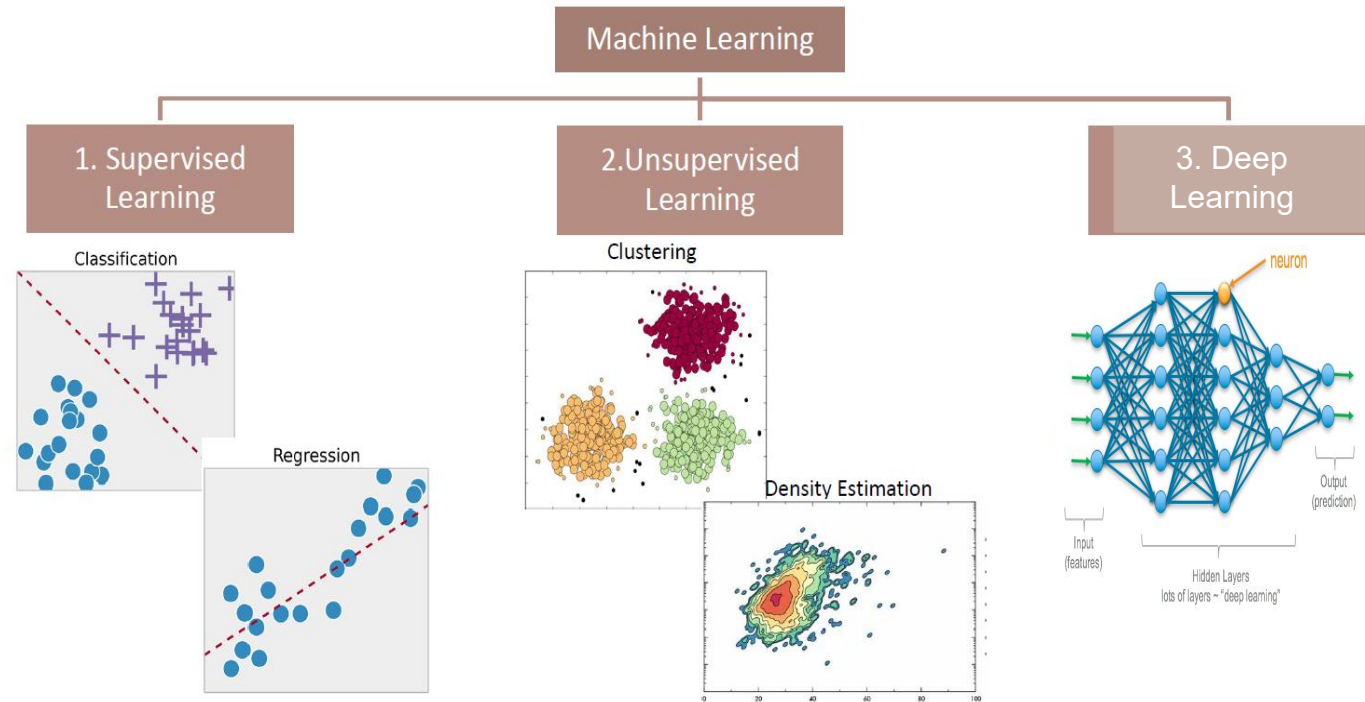


3. Deep Learning



Timeplan

- Lecture 1: Introduction to AI and ML
- Lecture 2: Python: Pandas – Data Handling.
- Lecture 3 & 4: Supervised ML - Regression.
- Lecture 5 & 6: Supervised ML - Classification.
- Lecture 7 & 8: Unsupervised ML.
- Lecture 9,10 & 11: Deep Learning.
- Lecture 12: Revision

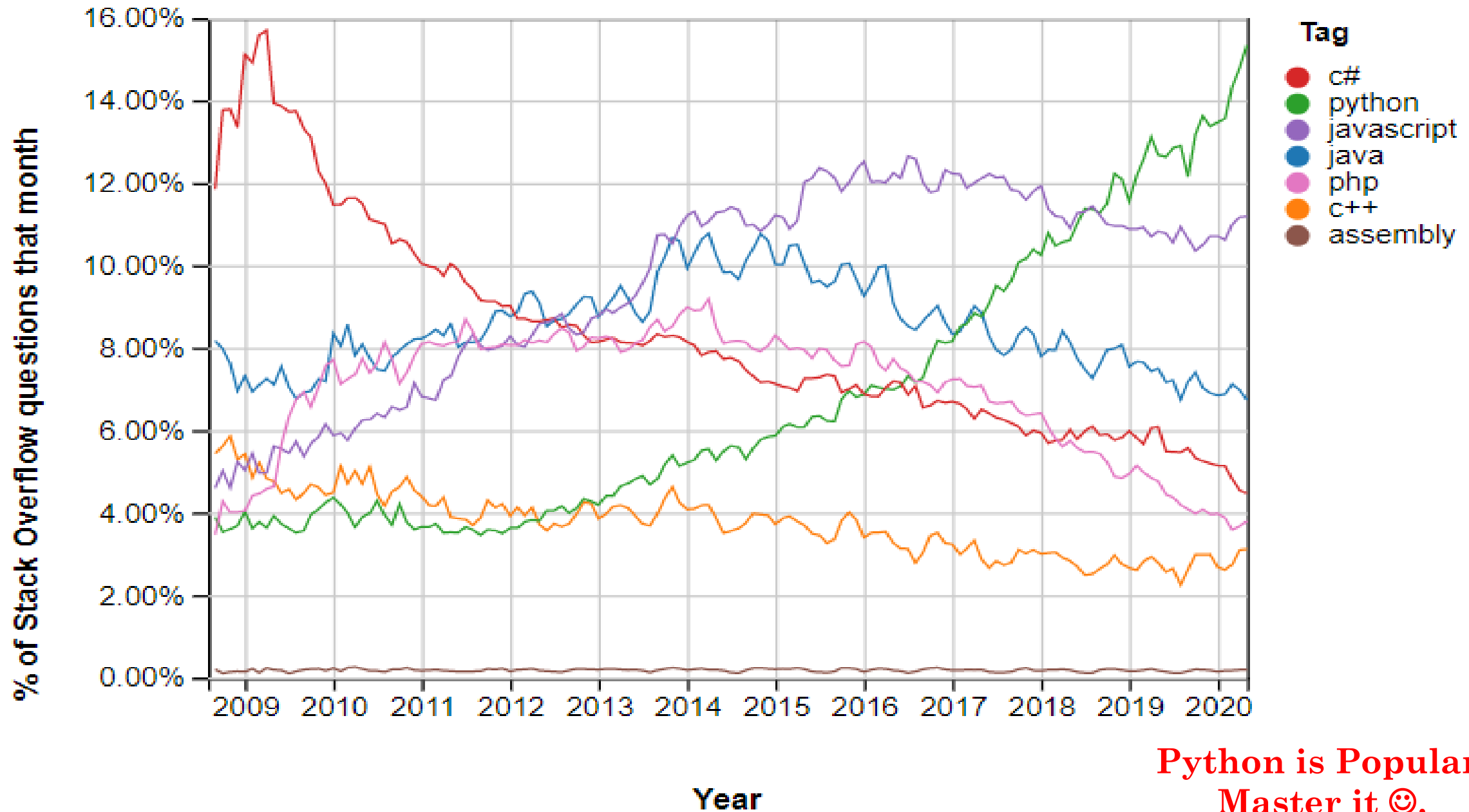


Programming Language



While **hundreds of languages** have been developed, **fewer than 30–40** account for the majority of real-world use in industry, academia, and software development today.

Language Popularity



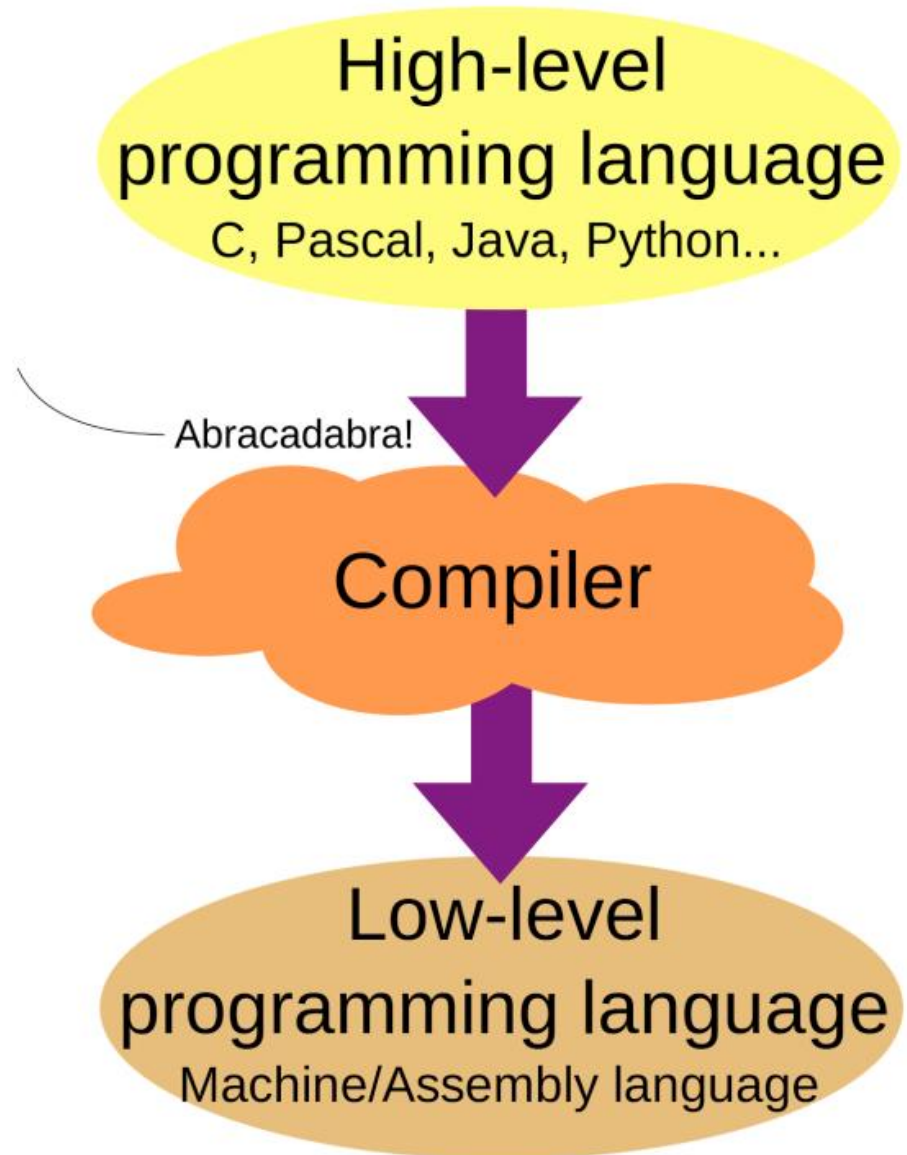
**Python is Popular,
Master it ☺.**

RedMonk

Image Source: <https://redmonk.com/>

Image Source: <https://redmonk.com/>

High Level vs. Low Level Programing Language



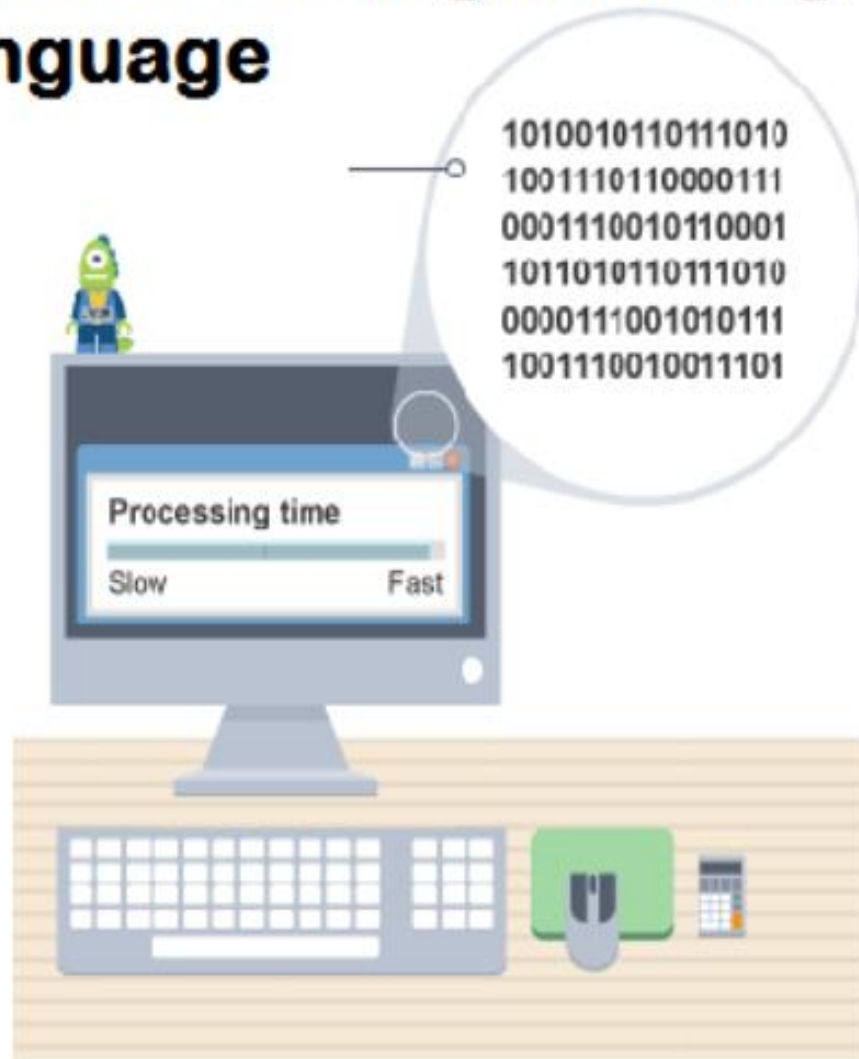
➤ Two Factors determine time to solution:

- How long it takes to write a program (human time)
- How long it takes that program to run (machine time)

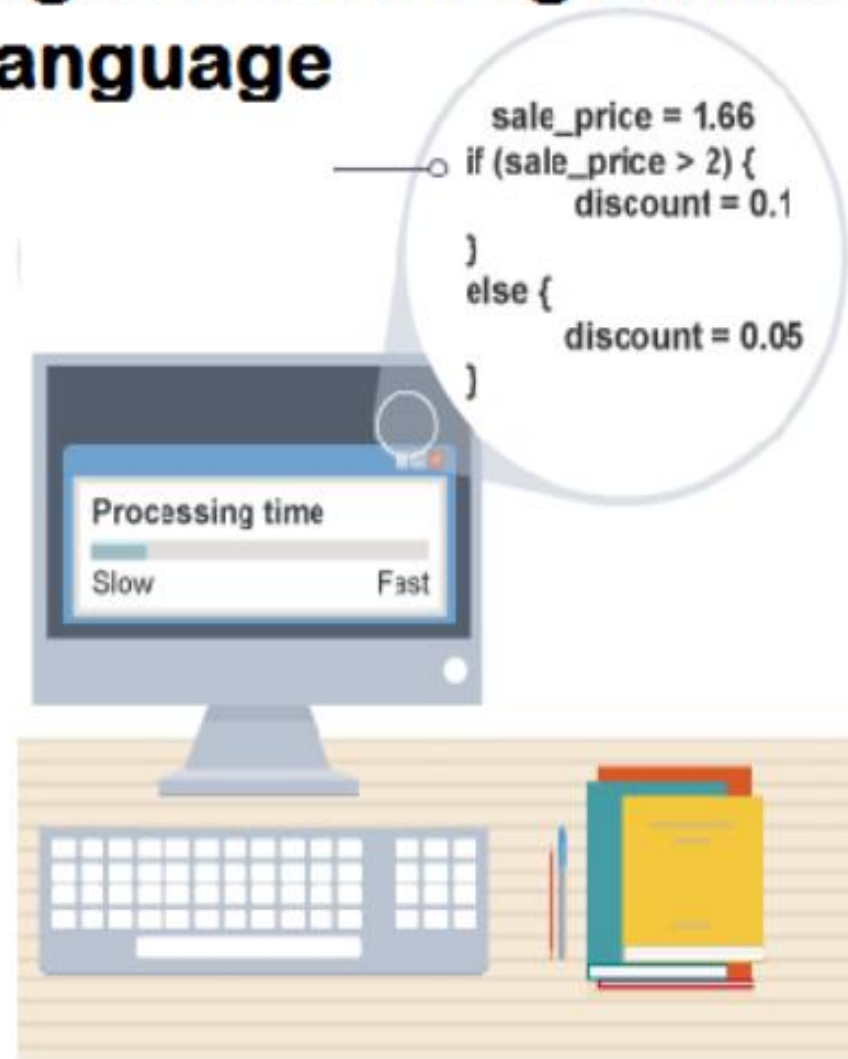
➤ Different languages make different tradeoffs between programming speed and execution speed.

- Programmers write the same number of lines of code per day no matter the language

Low Level Programming Language



High Level Programming Language



High Level vs. Low Level Programing Language

Feature	High-Level Language	Low-Level Language
Readability	Human-friendly	Machine-oriented
Speed	Slower	Faster
Control over Hardware	Limited	Full control
Memory Management	Mostly automatic	Manual
Portability	Cross-platform	Platform-specific
Examples	Python, Java, C#	Assembly, Machine Code

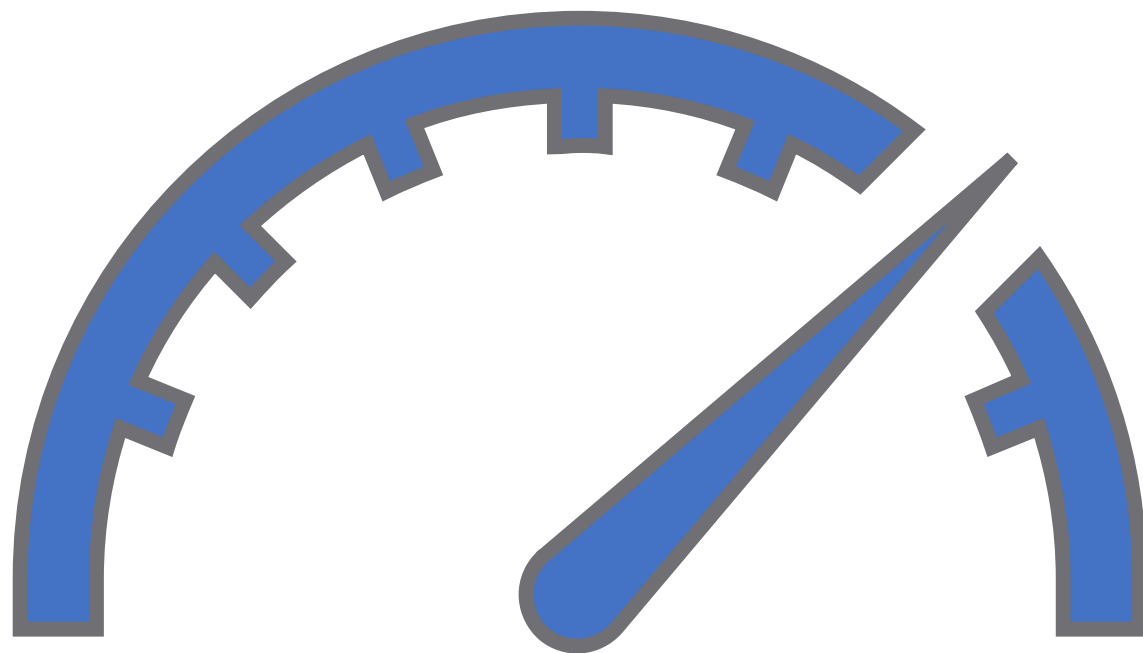
What is Python?

- Object-oriented, **high-level programming** language.
- Used as a scripting or glue language to connect existing components together.
- Simple, easy to learn syntax emphasizes readability.
- Supports modules and packages.
- Python interpreter and the extensive standard library are **FREE!**

Python creator

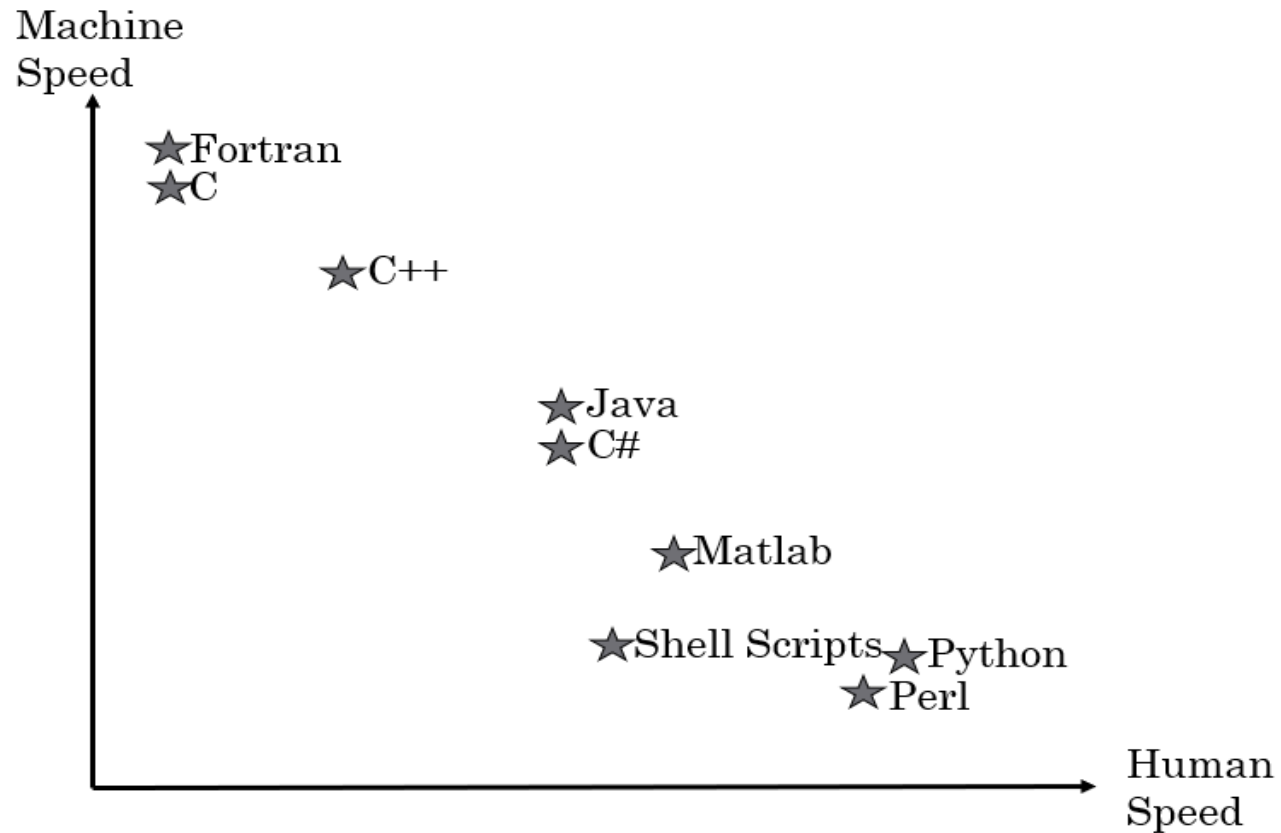
Guido van Rossum





Machine
Speed vs.
Human
Speed

Why Python?



Easy and Intuitive

Large Standard Library

Understandable, if you speak English

Active Open Source Community

Why Python?

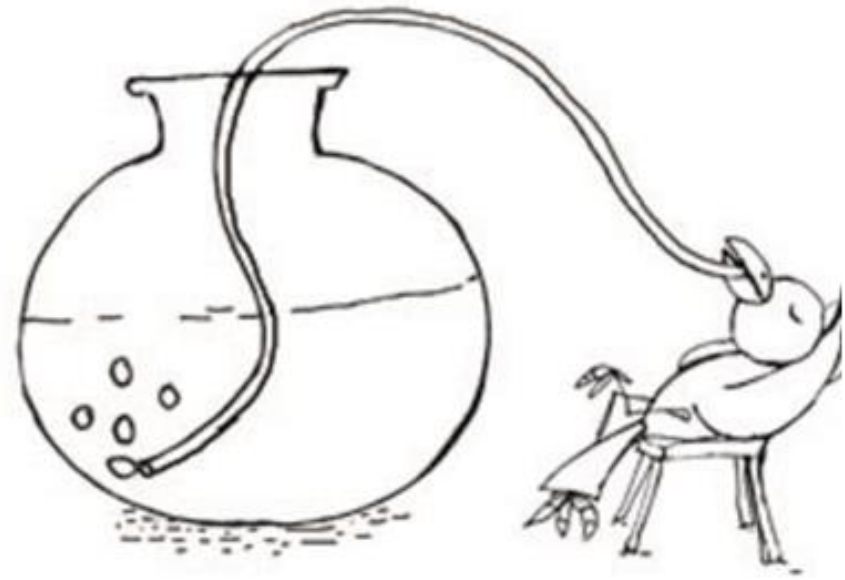
Non-programmer



Programmer



Python Programmer



Top 10 Python Development Trends for

2020

Constant Framework Upgrades

1

Python for Data Analysis and
Automation-Flexibility

2

Helping Hand for Cloud Computing
and Other Programming
Languages

3

Success in Academic

4

Huge Community Base around
the Globe

5

Easy Adaption

6

Smooth Integration with any
Programming languages

7

Free of cost and hence,
try-worthy

8

Hike in the Python users

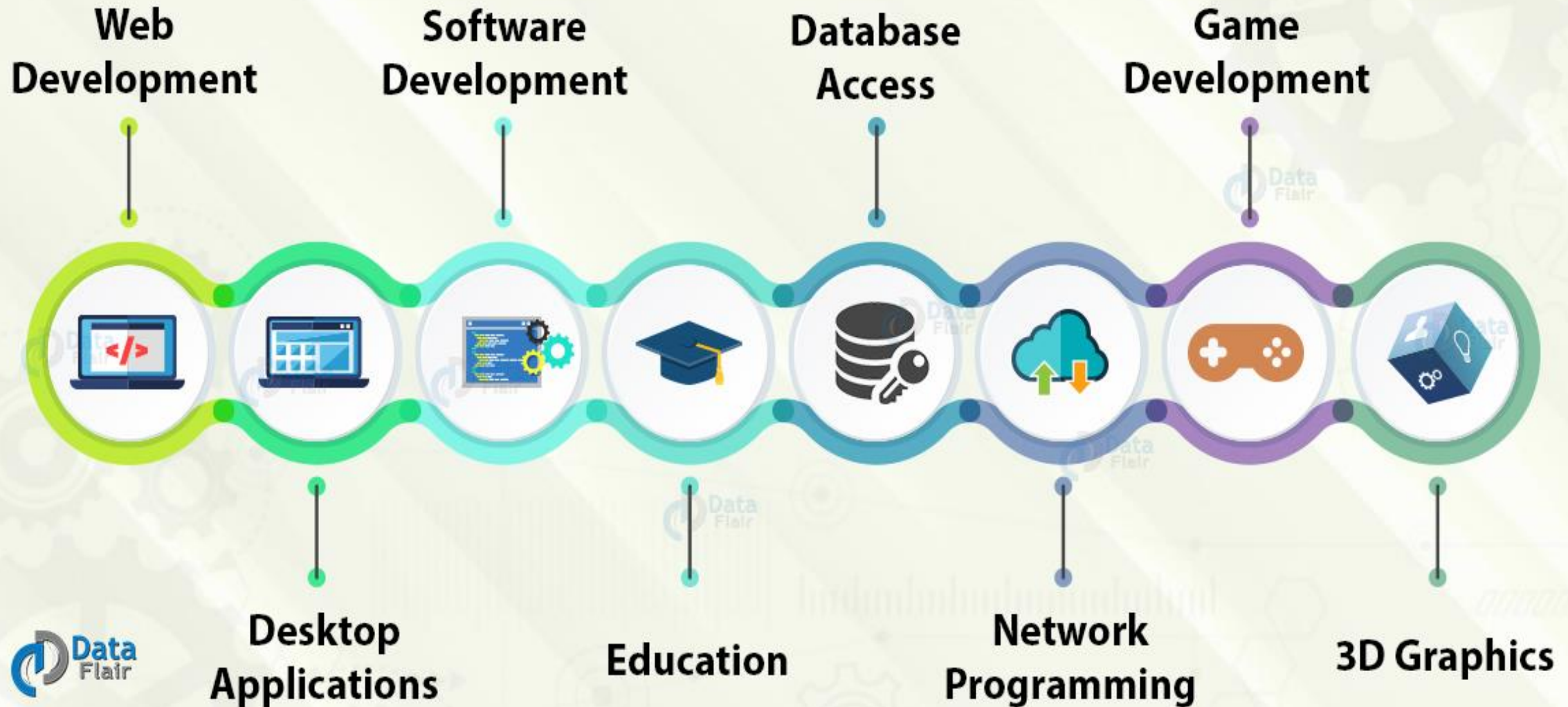
9

Multi-Functionality of Python

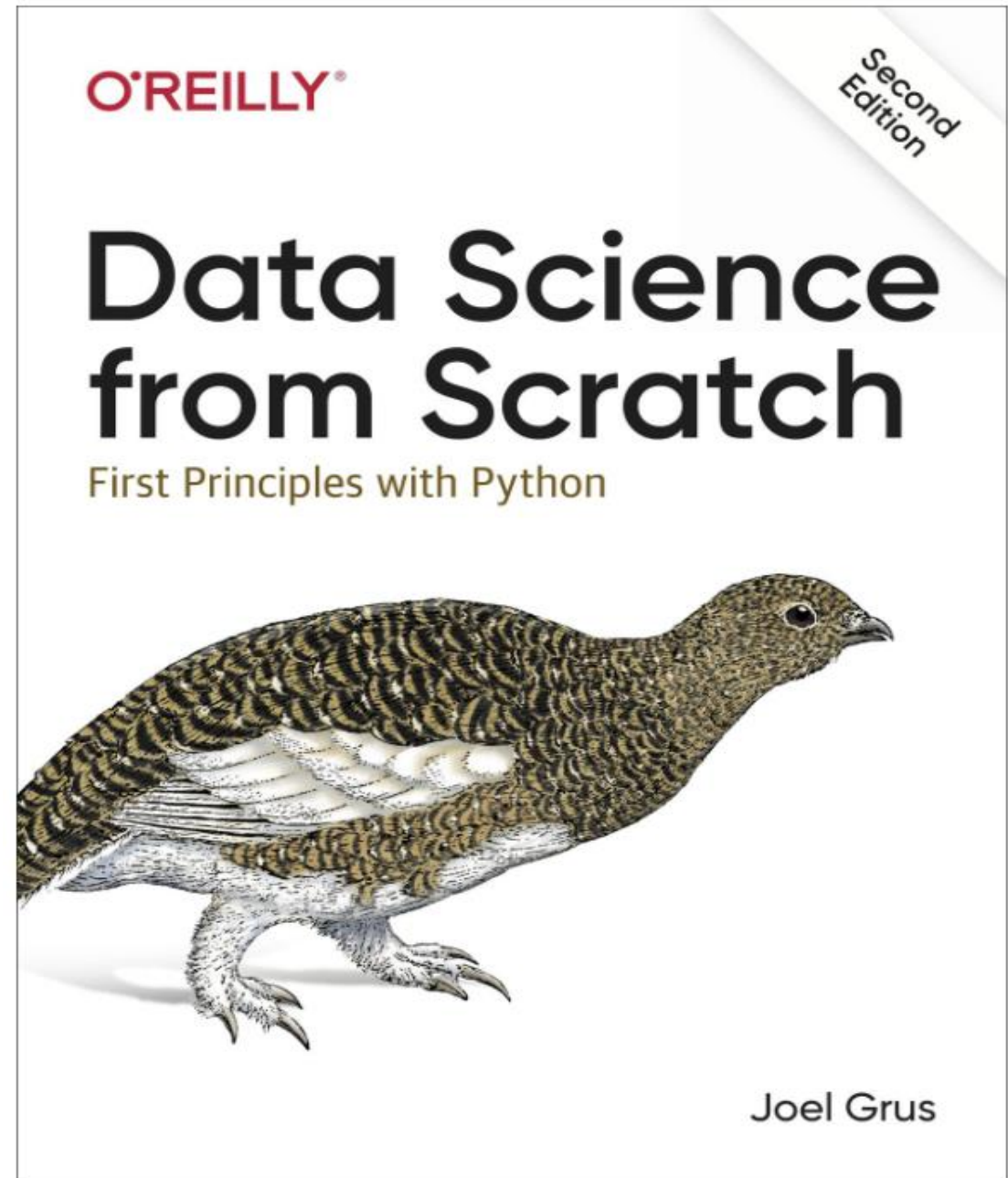
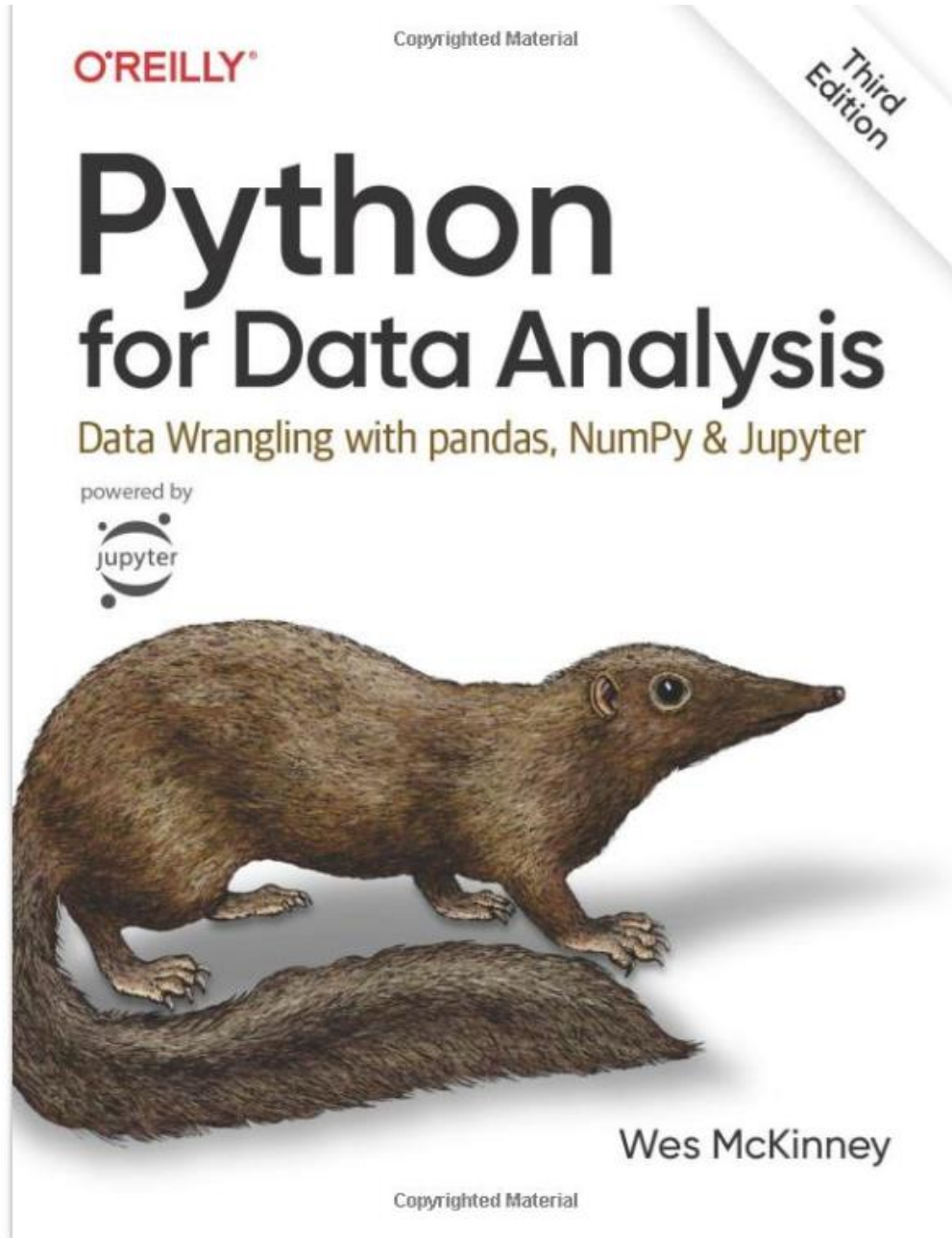
10



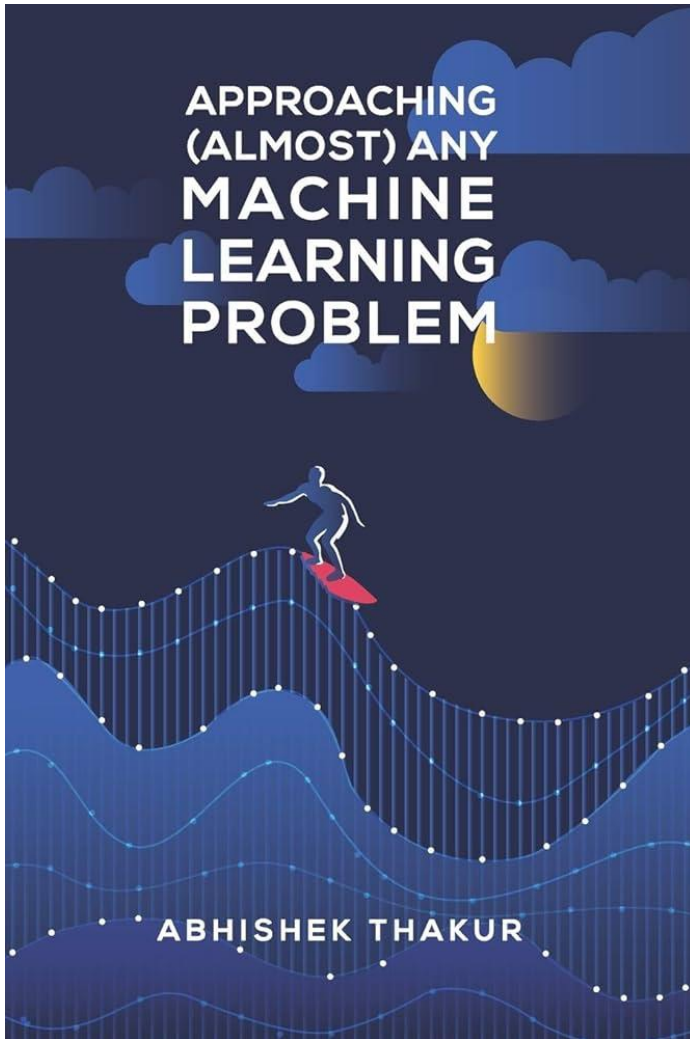
Python Applications



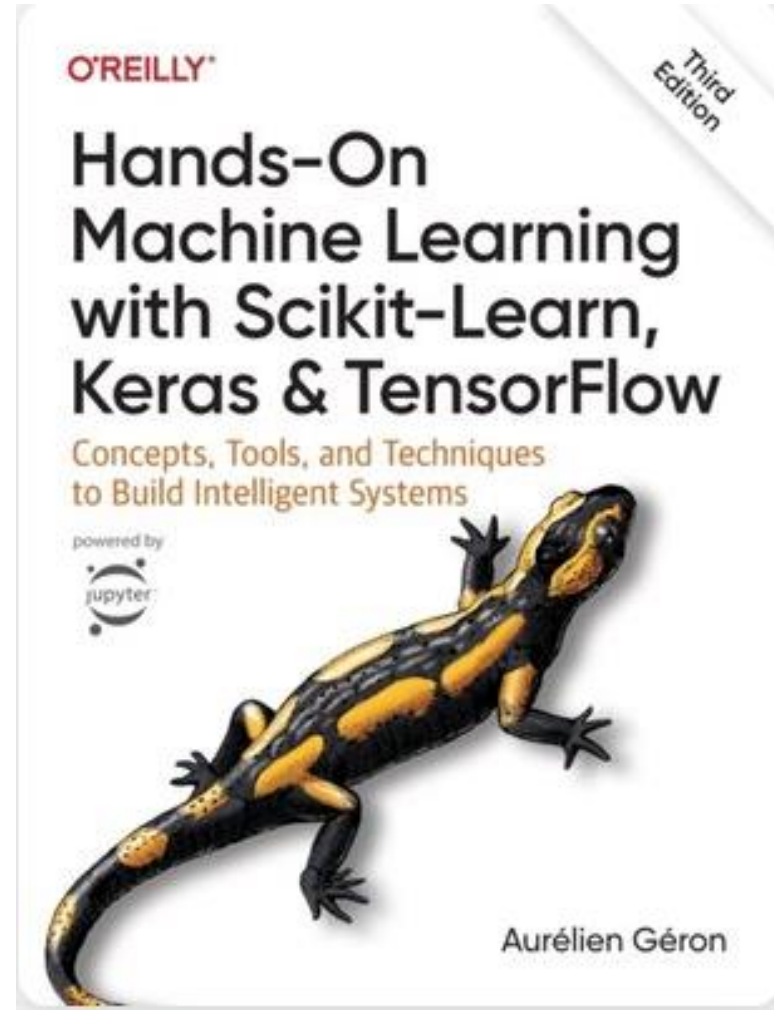
Useful Books for Python



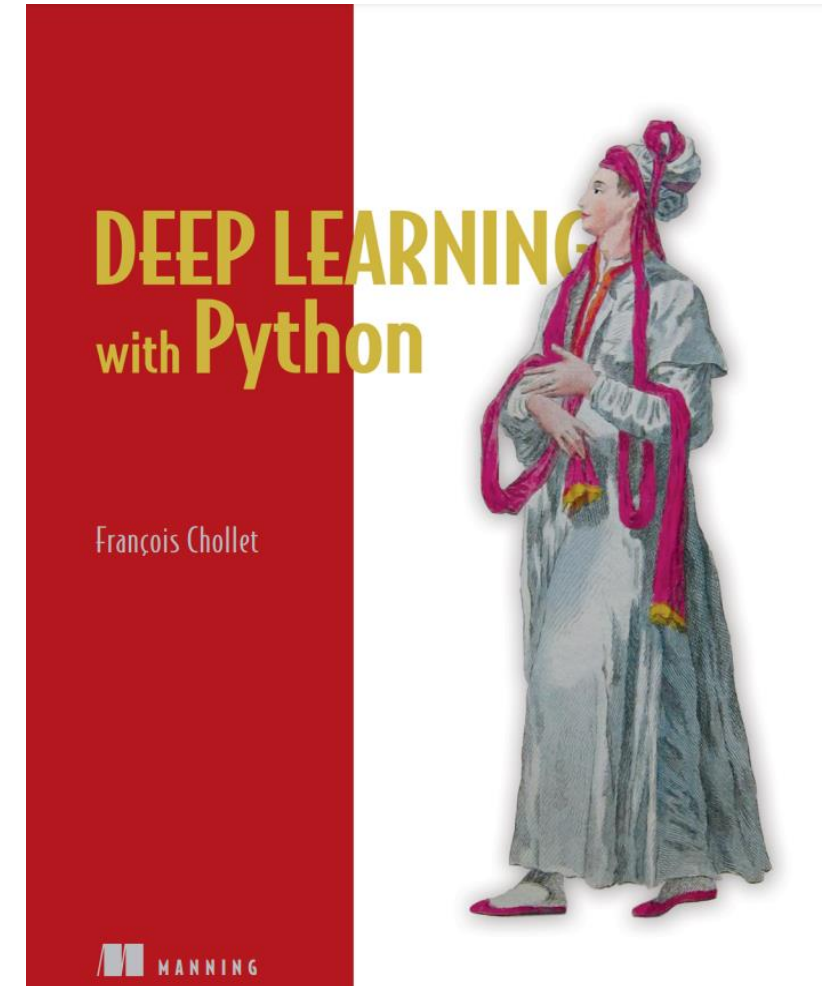
Reference Books for this Course



[Free PDF](#)

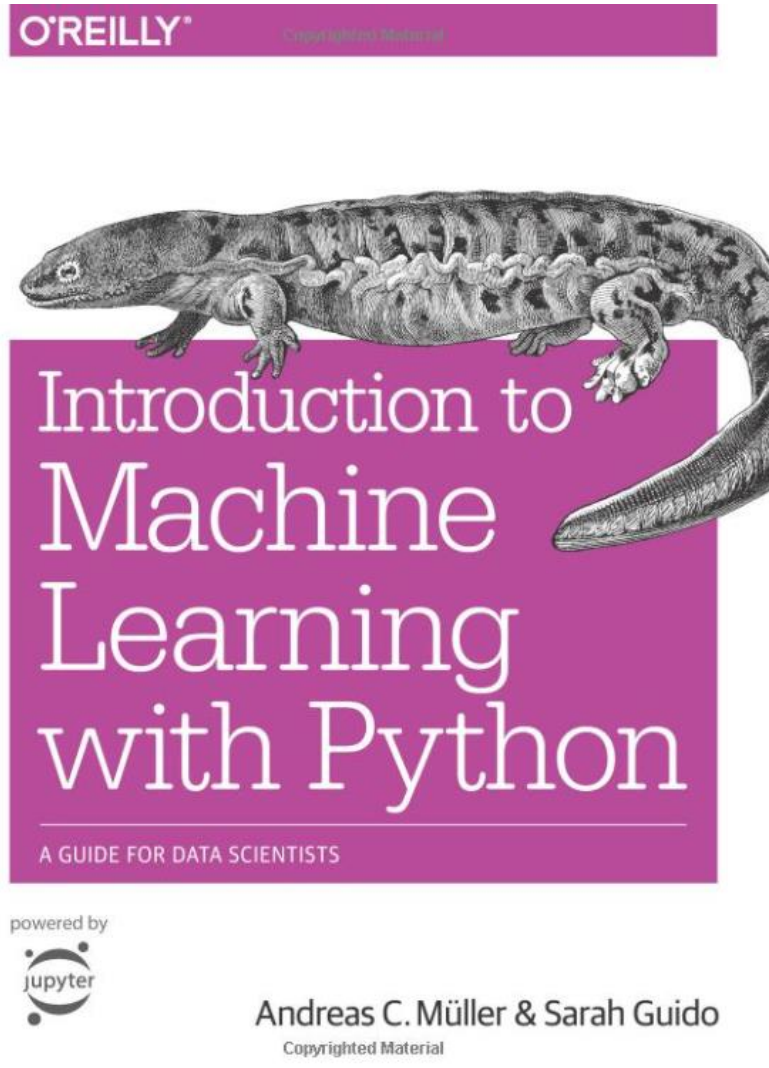


[BookSite](#)

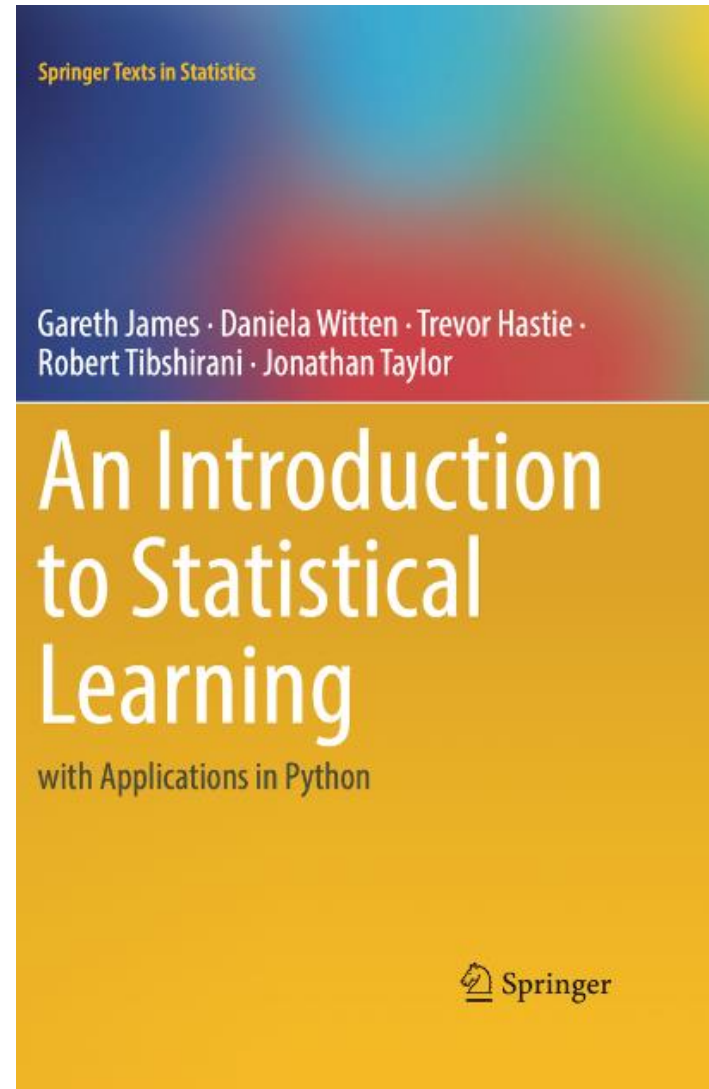


[Free PDF](#)

Other Useful Books for ML & DL



[Book Site](#)



[Free PDF](#)

Other Useful Resources

- Machine Learning- Tom Mitchell
- Machine Learning is Fun – Adam Geitgey
- Machine Learning: A Probabilistic Perspective – Kevin Murphy
- The Hundred Page ML book – Andriy Burkov

- Google AI Blog
- Facebook AI
- KG Nuggets
- Kaggle
- Medium-Data Science

<https://medium.com/swlh/top-20-websites-for-machine-learning-and-data-science-d0b113130068>

Top 20 Websites for Machine Learning and Data Science in 2020

Here is my list of the best websites for ML and data science to follow for valuable resources and news.



Benedict Neo [Follow](#)

Dec 10, 2019 · 13 min read ★



Installing Python

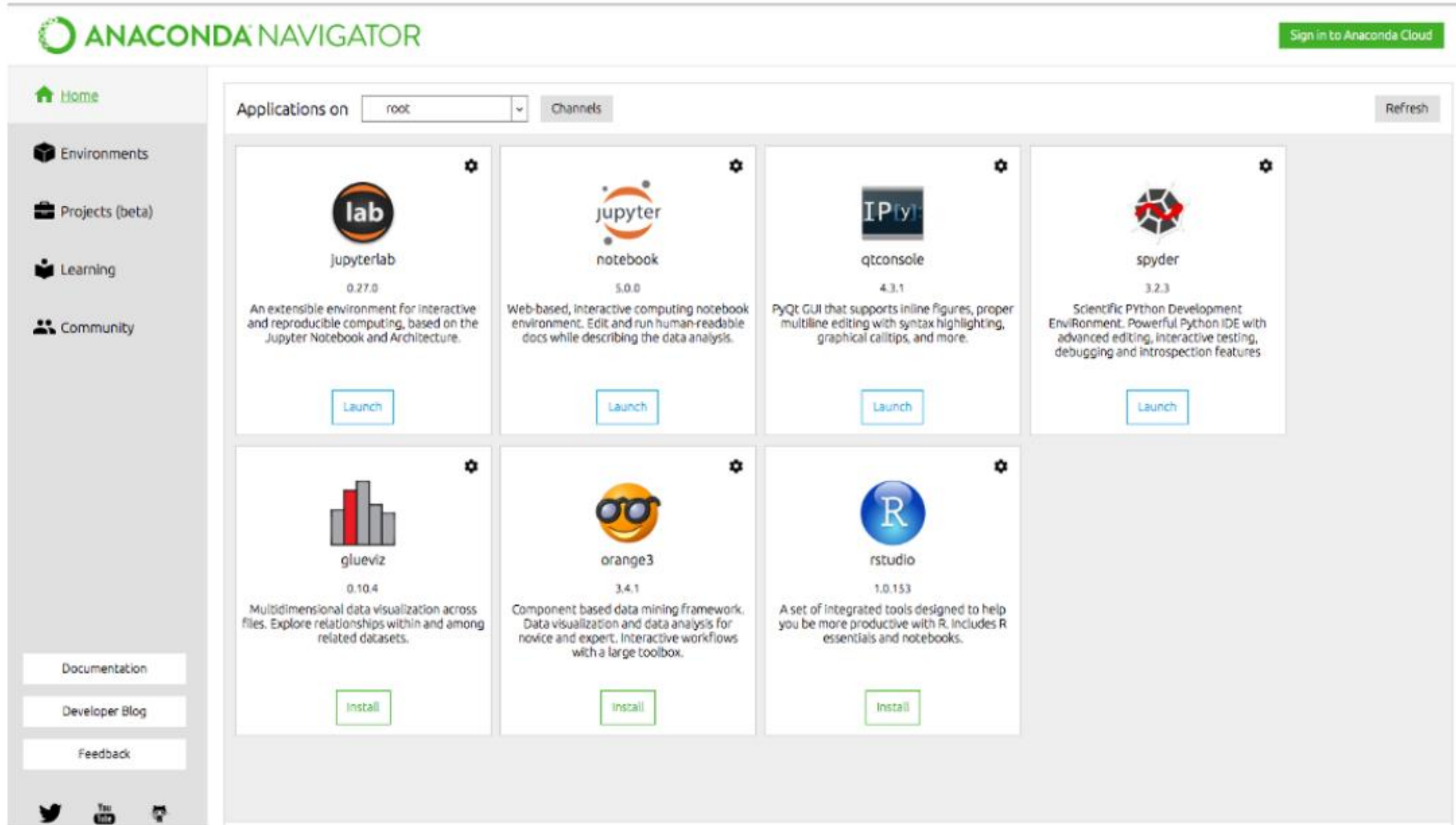
Python Development Environments

- **Anaconda/Spyder/Jupyter Notebook**
 - <https://www.anaconda.com/download/>



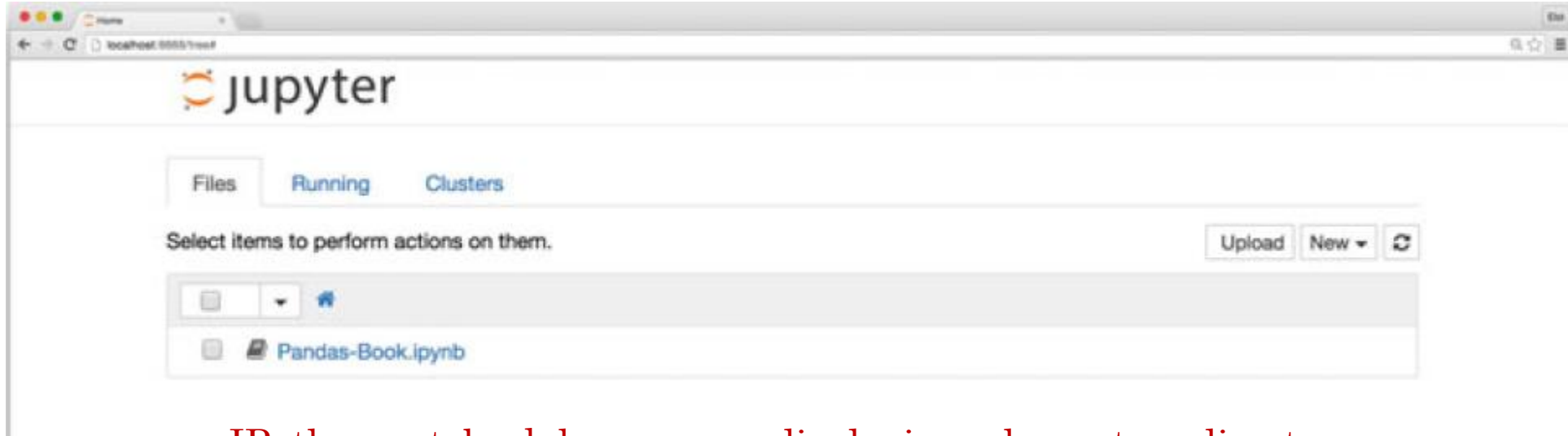
<https://medium.com/@GalarnykMichael/install-python-on-windows-anaconda-c63c7c3d1444>

Installing Python Using Anaconda

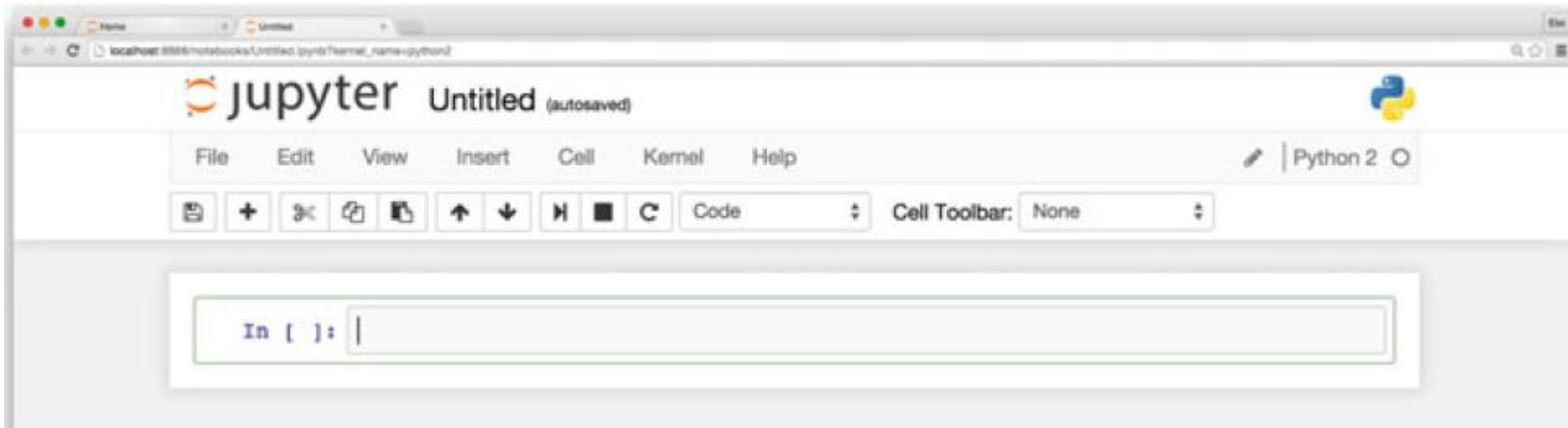


Anaconda Navigator Homepage(your screen might slightly defer in case of a newer version)

Launching Jupyter/IPython Notebook



IPython notebook home page, displaying a home tree directory



An empty new notebook

Online Python IDE

Click this: <https://jupyter.org/try>



Project Jupyter exists to develop open-source software, open-standards, and services for interactive computing across dozens of programming languages.

Online Python IDE

Click this: <https://deepnote.com/>

[Integrations](#)[Pricing](#)[For Teams](#)[For Education](#)[Join us](#)[Docs](#)[Log in](#)[Sign up](#)

Where data teams do their best work

Deepnote is a new kind of data science notebook. Jupyter-compatible with real-time collaboration and running in the cloud.

[Start using Deepnote now →](#)

Deepnote is completely free. Get started in 10 seconds.

Online Python IDE

Click this: <https://colab.research.google.com/>

 Welcome to Colaboratory

File Edit View Insert Runtime Tools Help

☰ Table of contents

🔍 Getting started

{x} Data science

🔑 Machine learning

📁 More resources

Featured examples

+ Section

+ Code + Text Copy to Drive

If you're already familiar with Colab, check out this video to learn about interactive tables, the executed code history view and the command palette.



[] Start coding or [generate](#) with AI.

What is Colab?

Colab, or 'Colaboratory', allows you to write and execute Python in your browser, with

- Zero configuration required
- Access to GPUs free of charge
- Easy sharing

Whether you're a **student**, a **data scientist** or an **AI researcher**, Colab can make your work easier. Watch [Introduction to Colab](#) to find out more, or just get started below!

▼ Getting started

The document that you are reading is not a static web page, but an interactive environment called a **Colab notebook** that lets you write and execute code.

For example, here is a **code cell** with a short Python script that computes a value, stores it in a variable and prints the result:

Practice for Python

Click this: <https://wiki.python.org/moin/FrontPage>



Search [titles](#) [text](#)

[» FrontPage](#)
[» FrontPage](#)

FRONTPAGE >>

RECENTCHANGES >>

FINDPAGE >>

HELPCONTENTS >>

Page

» Immutable Page

» Info

» Attachments

» More Actions: ▼

User

» Login

The Python Wiki

Welcome to the Python Wiki, a user-editable compendium of knowledge based around the Python programming language. Some pages are protected against casual editing - see [WikiEditingGuidelines](#) for more information about editing content.

 [Python](#) is a great object-oriented, interpreted, and interactive programming language. It is often [compared](#) (favorably of course 😊) to Lisp, Tcl, Perl, Ruby, C#, Visual Basic, Visual Fox Pro, Scheme or Java... and it's much more fun.

Python combines remarkable power with very clear syntax. It has modules, classes, exceptions, very high level dynamic data types, and dynamic typing. There are interfaces to many system calls and libraries, as well as to [various windowing systems](#). New built-in modules are easily written in C or C++ (or other languages, depending on the chosen [implementation](#)). Python is also usable as an extension language for [applications written in other languages](#) that need easy-to-use scripting or automation interfaces.

Getting Started	Events and Community	Software	Core Development	Using and Editing the Wiki
---------------------------------	--------------------------------------	--------------------------	----------------------------------	--

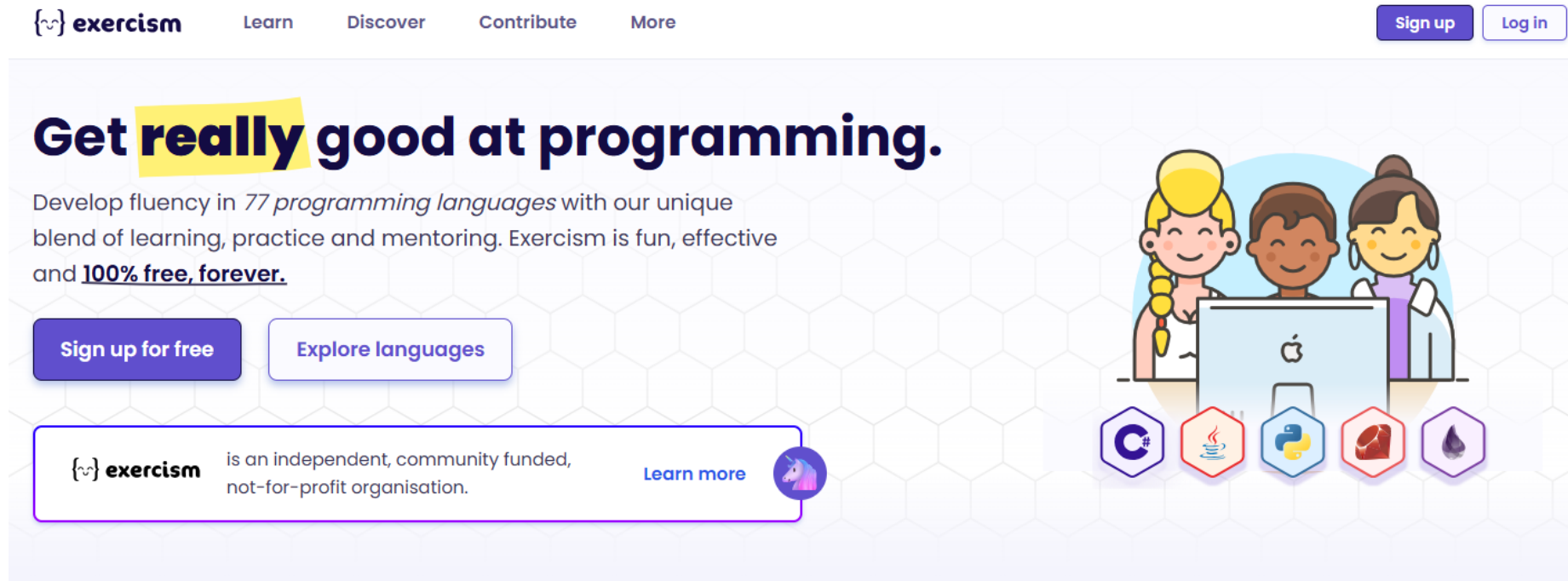
Getting Started

Beginners Guide	Documentation
Links to tutorials, courses and resources	Learning materials, topic guides and links to central resources
Beginner Errors	Python Books
Some common pitfalls of beginners	Books about Python plus reviews
Asking for Help	Python Audio Materials
Questions asked by beginners, answered here	A mixture of introductory and topical material
Languages	Python Implementations
Resources written in languages other than English	Different software which runs programs in the Python language

See also the [documentation category](#) for all known documentation-related pages.

Even More Practice for Python

Click this: <https://exercism.org/>

The image shows the top section of the Exercism website. At the top, there is a navigation bar with the Exercism logo (a curly brace with a tilde) and the word "exercism" in a bold, sans-serif font. To the right of the logo are links for "Learn", "Discover", "Contribute", and "More". Further right are two buttons: "Sign up" and "Log in". Below the navigation bar is a large hero section with a light blue background featuring a subtle hexagonal pattern. On the left side of the hero section, the text "Get **really** good at programming." is displayed, with "really" highlighted in a yellow box. Below this, a paragraph states: "Develop fluency in 77 programming languages with our unique blend of learning, practice and mentoring. Exercism is fun, effective and **100% free, forever.**" There are two buttons: "Sign up for free" and "Explore languages". Below these is a box containing the Exercism logo, the text "is an independent, community funded, not-for-profit organisation.", and a "Learn more" link with a small unicorn icon. On the right side of the hero section, there is an illustration of three diverse people (two women and one man) smiling and looking at a laptop. Below the laptop are five hexagonal icons representing different programming languages: C#, Java, Python, Ruby, and Go.

{~} **exercism** Learn Discover Contribute More [Sign up](#) [Log in](#)

Get **really** good at programming.

Develop fluency in 77 programming languages with our unique blend of learning, practice and mentoring. Exercism is fun, effective and **100% free, forever.**

[Sign up for free](#) [Explore languages](#)

{~} **exercism** is an independent, community funded, not-for-profit organisation. [Learn more](#)

The bottom of the page features several decorative icons. On the left, there is a green diamond, a double equals sign, and a zigzag line. In the center, there is a hexagon containing a grid pattern. On the right, there is a yellow square, a grid of dots, and a yellow triangle pointing downwards.

Explore and get fluent in
77 programming languages

Task 1



INSTALL ANACONDA



LAUNCHING
JUPYTER NOTEBOOK

Thank You
for
Attention

